

Sustained increase in flow-mediated dilation after daily intake of high-flavanol cocoa drink over 1 week.



A single-dose ingestion of flavanol-rich cocoa acutely reverses endothelial dysfunction. To investigate the time course of endothelial function during daily consumption of high-flavanol cocoa, we determined flow-mediated dilation (FMD) acutely (for up to 6 hours after single-dose ingestion) and chronically (administration for 7 days). The study population represented individuals with smoking-related endothelial dysfunction; in addition to FMD, plasma nitrite and nitrate were measured. The daily consumption of a flavanol-rich cocoa drink (3 x 306 mg flavanols/d) over 7 days (n=6) resulted in continual FMD increases at baseline (after overnight fast and before flavanol ingestion) and in sustained FMD augmentation at 2 hours after ingestion. Fasted FMD responses increased from 3.7 +/- 0.4% on day 1 to 5.2 +/- 0.6%, 6.1 +/- 0.6%, and 6.6 +/- 0.5% (each $P < 0.05$) on days 3, 5, and 8, respectively. FMD returned to 3.3 +/- 0.3% after a washout week of cocoa-free diet (day 15). Increases observed in circulating nitrite, but not in circulating nitrate, paralleled the observed FMD augmentations. The acute, single-dose consumption of cocoa drinks with 28 to 918 mg of flavanols led to dose-dependent increases in FMD and nitrite, with a maximal FMD at 2 hours after consumption. The dose to achieve a half-maximal FMD response was 616 mg (n=6). Generally applied biomarkers for oxidative stress (plasma, MDA, TEAC) and antioxidant status (plasma ascorbate, urate) remained unaffected by cocoa flavanol ingestion. The daily consumption of flavanol-rich cocoa has the potential to reverse endothelial dysfunction in a sustained and dose-dependent manner.